Huaiyuan QIN

Homepage · Email · Google Scholar · GitHub

Research Engineer, A*STAR, Singapore

Looking for PhD opportunities from Fall 2027, fully funded by the National Science Scholarship (PhD)

Research Interests

Can artificial learning become an empirical science, with discoverable laws of change and invariance? I study this question in vision models through data-centric AI approaches: how data shape what models learn, and how trained models can themselves serve as data.

Education

2019–2023 **Nanyang Technological University**, Singapore
B.Eng. (Highest Distinction) in EEE · GPA 4.93/5.00

Research Experience

2023–Present **Institute of Advanced Intelligence and Computing (IAIC), A*STAR**, Singapore
Research Engineer Supervisor: Dr. Hongyuan Zhu
Lead research projects on visual foundation models and efficient AI.

2021–2023 **Institute for Infocomm Research (I2R), A*STAR**, Singapore
Research Intern Supervisor: Dr. Hongyuan Zhu
Built an indoor disaster simulator in Unity and a 3D indoor scene graph generation method.

2021–2022 **Pensees Pte. Ltd.**, Singapore
AI Scientist Intern Supervisor: Dr. Jane Shengmei Shen
Developed 3D hand gesture and human pose estimation for monocular RGB and NIR cameras.

Publications

First author Listed in chronological order. · *: equal contribution. · †: project lead.

NeurIPS 2026
1. **Huaiyuan Qin**, Muli Yang, Gabriel James Goenawan, Peng Hu, Chen Gong, Xi Peng, Hongyuan Zhu
Attention Transfer Is Not Universally Effective for Vision Transformers
paper · project · code

CVPR 2026 Findings
2. **Huaiyuan Qin**, Muli Yang, Gabriel James Goenawan, Kai Wang, Zheng Wang, Peng Hu, Xi Peng, Hongyuan Zhu
Beyond Loss Values: Robust Dynamic Pruning via Loss Trajectory Alignment
paper · code

Preprint
3. **Huaiyuan Qin**, Muli Yang, Gabriel James Goenawan, Shiqi Huang, Min Kass Chong, Wahyu Wiratama, Peng Hu, Chen Gong, Wu Liu, Xi Peng, Chun Jian Ho, Hongyuan Zhu
Copy the Same, Distill the Difference: Initializing Linear Vision Transformers
paper · project · code

Preprint
4. **Huaiyuan Qin***, Gabriel James Goenawan*, Zihang Lin*, Muli Yang, Hongyuan Zhu
DiD It in 87 Minutes: A Label-Free Softmax-to-Linear Adaptation of Vision Transformers for Object Detection
paper · project

TMLR 2025
5. **Huaiyuan Qin**, Muli Yang, Siyuan Hu, Peng Hu, Yu Zhang, Chen Gong, Hongyuan Zhu
Beyond Instance Consistency: Investigating View Diversity in Self-supervised Learning
paper · project · code

Project lead

EMNLP 2026 Findings
6. Zihang Lin, **Huaiyuan Qin**†, Muli Yang, Hongyuan Zhu
SDGBiasBench: Benchmarking and Mitigating Vision–Language Models' Biases in Sustainable Development Goals
paper · code · data

Co-author

AAAI 2026

7. Muli Yang, Gabriel James Goenawan, Henan Wang, **Huaiyuan Qin**, Chenghao Xu, Yanhua Yang, Fen Fang, Ying Sun, Joo-Hwee Lim, Hongyuan Zhu
Your AI-Generated Image Detector Can Secretly Achieve SOTA Accuracy, If Calibrated
paper · code

ICCV 2025
Highlight

8. Shiqi Huang, Shuting He, **Huaiyuan Qin**, Bihan Wen
SCORE: Scene Context Matters in Open-Vocabulary Remote Sensing Instance Segmentation
paper · code

CVPR 2025

9. Muli Yang, Gabriel James Goenawan, **Huaiyuan Qin**, Kai Han, Xi Peng, Yanhua Yang, Hongyuan Zhu
Detecting Open World Objects via Partial Attribute Assignment
paper · code

ICIPW 2024
AI4IPoT
Best Paper

10. Muli Yang, **Huaiyuan Qin**, Hongyuan Zhu
Dynamic Task-Oriented Prompting for Generalized Aerial Category Discovery
paper

Patents and Technology Disclosures

2026 (TD) *A Robust Framework for AI-Based Image Enhancement.*
2026 (TD) *A Fast Framework for AI-Based Image Enhancement.*
2025 (SG Patent) *A System for Generating a Scene Knowledge Graph for a High-resolution Image.*
2025 (TD) *STAR-KG: Scale-aware TrAnsformeR for scene Knowledge Graph in Satellite Imagery.*
2024 (TD) *Discovering Novel Remote Sensing Categories via Adaptive Task-oriented Prompting.*

Mentoring

2026–Present **Zonghan Du**, PhD student at USTC
2025–Present **Zihang Lin**, undergraduate at NTU; now a research intern at UIUC
2023–2025 **Siyuan Hu**, undergraduate at NTU; now a PhD student at MMLab@NTU

Honors and Awards

2026 **Top Reviewer**, NeurIPS 2026
2025 **National Science Scholarship (PhD)**, A*STAR
2024 **Best Paper Award**, ICIP AI4IPoT Workshop 2024
2019–2023 **Dean's List**, Academic Years 2019–20, 2020–21 and 2022–23
2020–2022 **NTU President Research Scholar**, Academic Years 2020–21 and 2021–22
2018 **NTU Science and Technology Undergraduate Full Scholarship**

Service and Outreach

Reviewer NeurIPS, ICLR, ECCV, ACL, AAAI, IJCAI, ACM MM, ICIP, IEEE TIP
Volunteer Main committee for **Purple Parade 2019**, Singapore's annual movement to promote awareness and celebrate the abilities of persons with disabilities.

[CV compiled on October 20, 2026]